Rajan Patkar

rajpatkar.github.io/contact (309) 863-7053 rajan.patkar.email@gmail.com

Highlights:

- Master's student at the University of Illinois seeking a full-time job starting after 2027/05
- Pursuing research in cybersecurity for critical infrastructure & cyberphysical systems
- Continue to hold current Secret-level clearance, former Scholarship for Service recipient

Experience:

Intern in Energy Systems at MIT Lincoln Laboratory (2026/05 - 2026/08):

- Owned project on converting OT inventories into high-fidelity network representations
- Investigated disparate installation docs to construct comprehensive NetBox OT inventory
- Devised novel pipeline from NetBox data to interactive GNS3 OT network simulation
- Demonstrated utility of simulation and proposed rehosting OT firmware on QEMU VMs

Research Assistant (prev. Intern) at the University of Illinois (2026/08 - now, 2025/05 - 2026/05):

- Targeting improving security and reliability within operational technology (OT) networks
- Worked on project finding and fixing vulnerabilities in OT devices on distributed networks
- Evaluated DPU performance in monitoring kernel health of network of smart devices
- Consulted on implementation of CIE practices to mitigate simulated OT network attacks

Cybersecurity Intern at the University of Illinois Information Trust Institute (2024/05 - 2024/08):

- Crafted hardware/software suite to determine efficiency of large-scale SSL offloading
- Designed network agents in C to record packet timing data for network performance
- Utilized Python automation package to deploy network agents and run experiments
- Presented experiment findings to recommend OT networks continue with on-device SSL

Engineering Intern/Student Trainee at Caterpillar (2022/05 - 2022/08, 2021/06 - 2021/08):

- Automated collection and processing of machine data reports from database with Python
- Identified serial data communication problems (CAN + J1939) on engine control modules
- Automated testing of calibration software for hydraulic excavators using MATLAB

Education:

M.S. in Electrical and Computer Engineering at the University of Illinois Urbana-Champaign

- Prospective graduation and thesis submission 2027/05, currently have a 3.69 GPA
- Continuing from B.S. in Computer Engineering at Illinois (grad. 2025/05) - 3.72 GPA
- During B.S., study abroad semester (fall 2024) at Hanyang University in Seoul, SK

Skills:

- Proficient in C, C++, x86/RISC assembly, Python, MATLAB, R, Java, JS, and ArcGIS
- Adept interpersonal communicator (native English/Marathi, B2 Spanish/Korean)